Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
5	*(a)*	(i)		D		1		1		1
		(ii)		[k = 0.17 × 1 000 =] 170 [N/m]		1		1	1	1
	(b)	(i)		$F = kx = 120 \times (10.7 \times 10^{-3}) = 1.284$ [N] (1) Substitution into: $E = ½ Fx$ i.e. $E = 0.5 \times 1.284$ **ecf** $\times (10.7 \times 10^{-3})$ **ecf** (1) = 0.0068694 [J] (1) Answer of 6.9×10^{n} where n is not -3 award a maximum of 2 marks	1	1 1		3	3	3
		(ii)		Substitution into PE = mgh i.e. 0.0068694 **ecf** = $0.015 \times 10 \times h$ (1) $h = \frac{0.0068694}{0.015 \times 10}$ (1-rearrangement) h = 0.045796 [m] (1) Accept 0.0458 or 0.046 or 0.05 Answer of 4.6×10^{n} where n is not -2 award a maximum of 2 marks Award 3 marks for an answer of 8.56 with **ecf** applied for E = 1.284 [J] Answer of 8.56×10^{n} where n is not 0 award a maximum of 2 marks	1	1 1		3	3	3

Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
		(iii)	I	Dist = area under graph = $0.5 \times 0.9 \times 0.09$ (1) = 0.0405 [m] which is different (1) Accept 0.041 [m]			2	2	2	2
			II	Some of the stored energy in the spring is lost to heat **or** work done (1) As the spring retracts **or** friction **or** air resistance (1)	2			2		2
			III	Choose a spring that has a bigger stiffness constant **or** stiffer spring / bigger [maximum] extension **or** longer spring / smaller mass **or** weight ball / use spring D or F		1		1		1
				Question 5 total	**4**	**7**	**2**	**13**	**9**	**13**